Trade-off Between Accuracy and Computation in Decision Tree and Random Forest

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GitHub:Individual-Assignment-Machine-Learning-Tutorial.git

1. Introduction:

There is a constant trade-off between model accuracy and computational efficiency. Decision Trees and Random Forests are two popular supervised learning algorithms that illustrate this trade-off clearly and are widely used for both classification and regression problems. A Decision Tree is a single, hierarchical structure that recursively splits the data into smaller, more homogeneous groups based on feature values, producing a set of human-readable rules that are easy to interpret. In contrast, a Random Forest is an ensemble of many decision trees trained on bootstrapped samples and random subsets of features, and it combines their individual predictions to obtain a more robust final output.

This project investigates how these two algorithms perform on the Breast Cancer Wisconsin dataset using scikit-learn, with a particular focus on the balance between predictive performance and computational cost. Both models are trained on standardized features with an 80/20 train–test split to ensure a fair comparison. For each model, we measure test accuracy and error rate, as well as training and prediction time, and we also approximate model complexity using tree depth, number of nodes, and serialized model size in bytes. The Decision Tree provides good accuracy with very fast training and prediction, and a compact model that is easy to understand. However, the Random Forest achieves higher accuracy and lower error at the cost of significantly longer training time and a much larger model size. By analysing these results, the report aims to highlight how the choice between Decision Tree and Random Forest depends on the desired balance between accuracy, speed, and resource usage in real-world applications.

2. Key differences between Decision Tree and Random Forest:

2.1. Model structure:

A Decision Tree is a single tree that keeps splitting the data based on feature values until it reaches pure or nearly pure leaves. It produces one path of rules from root to leaf for each prediction. In contrast, a Random Forest builds many different decision trees on slightly different versions of the data and combines their predictions, usually by majority vote.

2.2. Bias–variance / overfitting:

A single Decision Tree has low bias but high variance, meaning it can fit the training data very well but is sensitive to noise and small changes in the dataset. This makes it more likely to overfit, especially if the tree is deep. A Random Forest keeps a similar bias but greatly reduces variance because it averages many trees, so individual overfitting errors tend to cancel out.

2.3. Data usage:

In a Decision Tree, each split is chosen using all training samples that reach that node and all available features. In a Random Forest, each tree is trained on a bootstrap sample (random sample with replacement) of the training data, and at each split the tree only considers a random subset of features. This extra randomness makes the trees more diverse and the ensemble more robust.

2.4. Performance:

Decision Trees can be unstable: a small change in the data can change the structure of the tree a lot and affect test accuracy. Random Forests are more stable and, as in your notebook, usually give higher test accuracy on the same dataset (0.956 vs 0.912 on the breast cancer data).

2.5. Interpretability and computation:

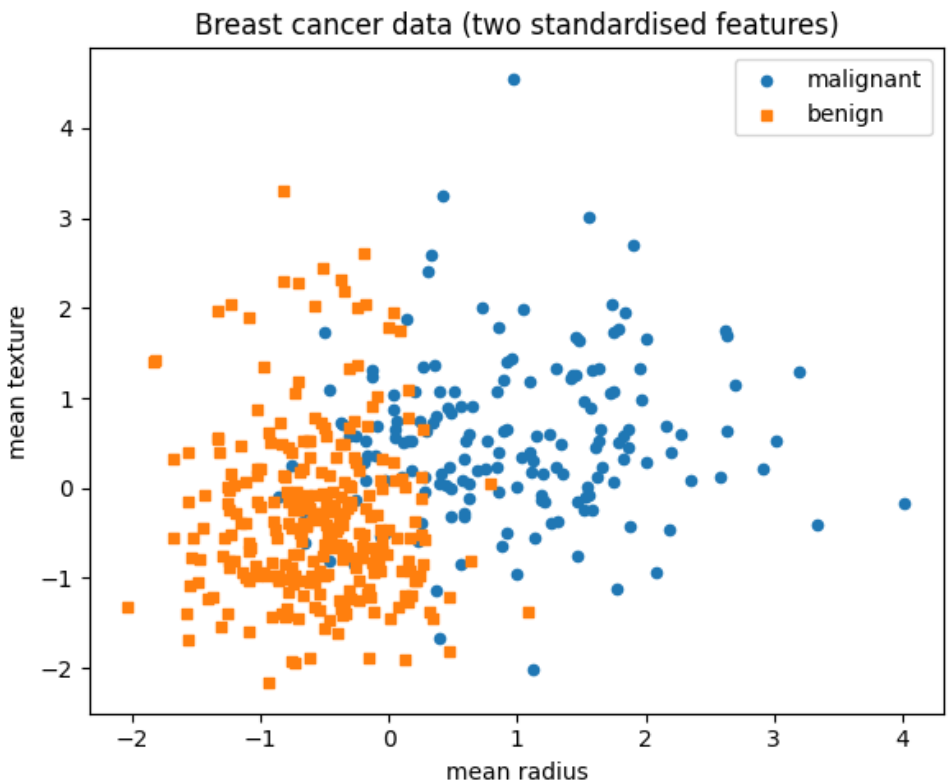
Decision Trees are easy to visualize and explain because you can follow a single path of decisions. Random Forests are harder to interpret directly; you typically rely on feature importance scores. They also need more computation and memory: in your experiment the Random Forest was slower to train and much larger in size than the single Decision Tree.

3. Methods:

3.1. Dataset:

We used the Breast Cancer Wisconsin dataset from scikit-learn, which contains 569 samples with 30 numeric features describing characteristics of cell nuclei in breast mass images. The target variable is binary, indicating whether a tumour is malignant or benign, with 212 malignant and 357 benign cases. This dataset is widely used as a benchmark for comparing classification algorithms in medical diagnosis tasks.

Scatter plot of two standardised features from the breast cancer dataset, with malignant and benign tumours shown using different colours and marker shapes. This visualisation helps illustrate how the classes overlap in feature space and why tree-based models like Decision Trees and Random Forests can separate them using axis-aligned splits.



3.2. Data preprocessing:

The dataset was first loaded into memory and split into features X and labels y. A stratified train–test split was applied with 80% of the data used for training and 20% for testing, using a fixed random state of 42 to ensure reproducibility. Because the features are on different scales, a StandardScaler was fitted on the training data and then used to transform both the training and test sets, so that each feature has zero mean and unit variance.

3.3. Model training:

Two supervised classification models were trained: a Decision Tree classifier and a Random Forest classifier from scikit-learn. Both models were fitted on the standardized training data using the same random state of 42 for fair comparison. The Random Forest used 200 trees with no explicit maximum depth, allowing the ensemble to learn complex patterns while reducing variance through bagging and feature randomness.

3.4. Evaluation and measurements:

Model performance was evaluated on the standardized test set using accuracy and the full classification report (precision, recall, and F1-score) for both malignant and benign classes. Confusion matrices were plotted to visualise the distribution of correct and incorrect predictions for each model. To study the trade-off between accuracy and computation, training time and

prediction time were measured separately for each model, and approximate model complexity was estimated using tree depth, number of nodes, and serialized model size in bytes.

4. Model results:

4.1. Model performance:

- The Random Forest clearly performs better than the single Decision Tree.
  - Decision Tree accuracy: 0.912
  - Random Forest accuracy: 0.956
    This means the ensemble model makes fewer classification mistakes overall.
- In terms of class-wise metrics, Random Forest improves both malignant and benign detection:
  - Decision Tree: malignant f1=0.89, benign f1=0.93
  - Random Forest: malignant f1=0.94, benign f1=0.97
    So it is better at catching cancers (malignant) and avoiding false alarms (benign).

4.2. Computation vs accuracy trade-off:

- The Decision Tree is much lighter computationally:
  - Train time ≈ 0.011 s, predict time ≈ 0.0006 s, model size ≈ 4 KB
- The Random Forest is far more expensive:
  - Train time ≈ 0.372 s, predict time ≈ 0.013 s, model size ≈ 635 KB

So, we gain about **4–5%** extra accuracy but pay with:

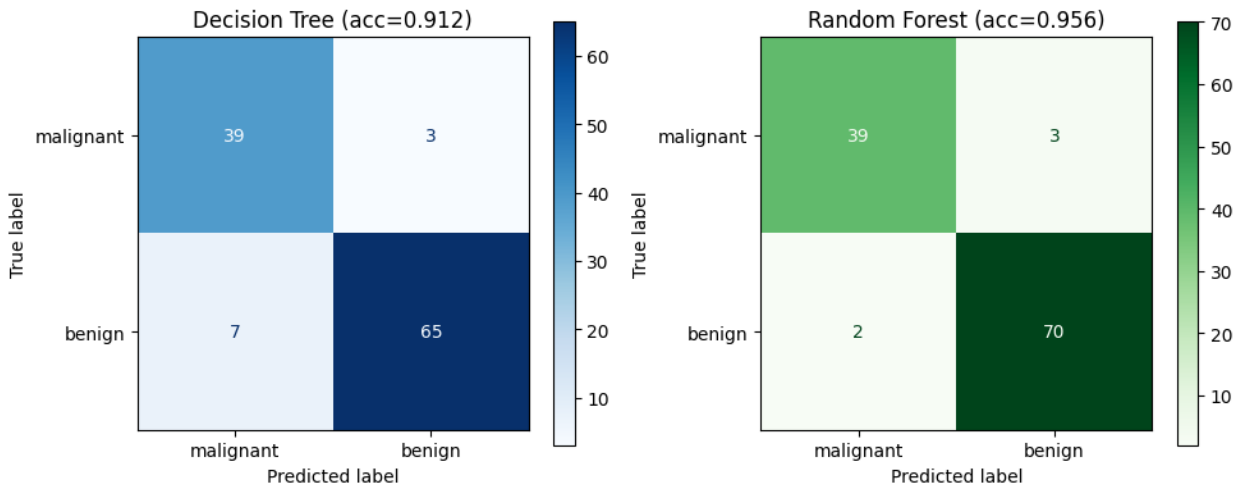
- ~30× slower training
- ~20× slower prediction
- ~150× larger model size

4.3. Table:

Model	Accuracy	Train time (s)	Predict time (s)	Size (bytes)
Decision Tree	0.912	0.0114	0.0006	4,104
Random Forest	0.956	0.3718	0.0126	634,547

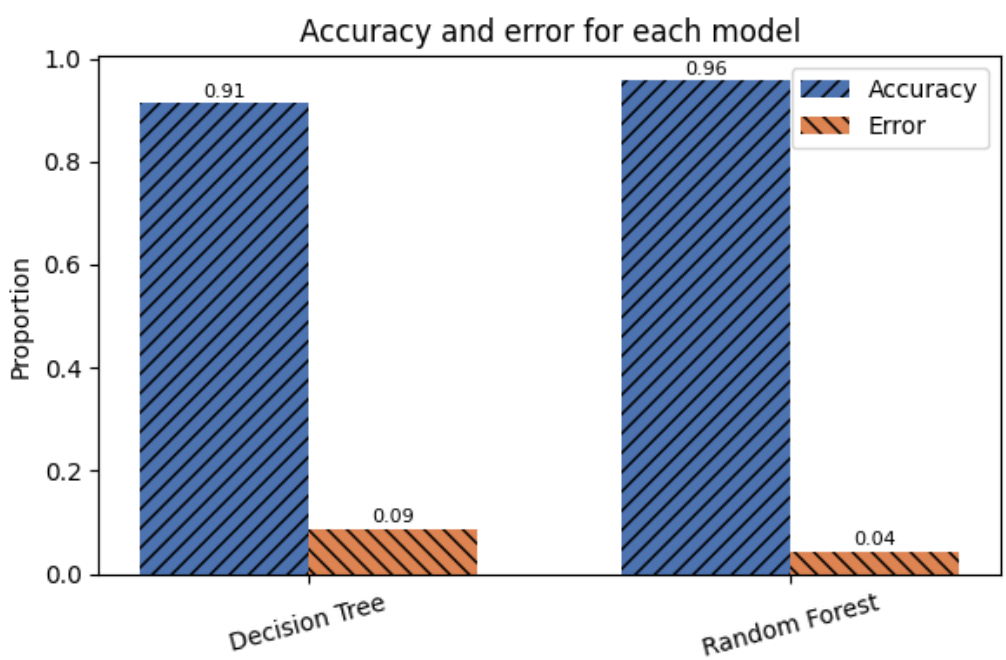
4.4. Visual results interpretation:

These confusion matrices can be used in teaching to show that a higher diagonal (and fewer off-diagonal entries) directly corresponds to higher accuracy and F1-scores, and to highlight which types of errors (missed cancers vs false alarms) each model makes.



Key interpretation

- Both models are equally good at identifying malignant cases on this test split (same 3 misses), but Random Forest is better at correctly recognising benign tumors.
- This explains why overall accuracy and benign F1-score are higher for Random Forest, while also reducing patient anxiety from false cancer alarms



The bar chart is a simple teaching tool: students can immediately see that Random Forest's higher green bar (accuracy) and lower red bar (error) come at the cost of more computation, reinforcing the trade-off visually rather than just with numbers.

- For Decision Tree, the blue bar is accuracy ≈ 0.91, and the orange bar is error ≈ 0.09.
- For Random Forest, the blue bar is higher at ≈ 0.96, and the orange error bar is lower at ≈ 0.04.

5. Does a Random Forest always have higher accuracy than a single Decision Tree on the same dataset?

No. Random forests often, but not always, have higher accuracy than a single decision tree on the same dataset.

5.1. Typical pattern:

- In many real-world tasks, RF wins because bagging many trees reduces variance and overfitting, so test accuracy and AUC are usually higher than a single tree.
- Textbook comparisons and tutorials usually show RF > DT on test accuracy, which is what you're seeing on the breast cancer dataset.

5.2. When a decision tree can be better:

- Very small or very simple datasets: a shallow tree can already capture the pattern; an RF may overfit noise or not gain much, so test accuracy can be similar or sometimes lower, especially with bad hyperparameters.
- Poor RF settings: too few trees, wrong max_depth, no class-imbalance handling, or bad feature subsets can make RF underperform a reasonably tuned single tree on that dataset.
- Different metric: you might care about recalling a minority class or interpretability instead of raw accuracy; a carefully pruned tree can be preferable even if RF has slightly higher accuracy.

6. Conclusion:

- Random Forest achieved higher accuracy and lower error than the Decision Tree on the breast cancer dataset, but required much more training time, prediction time, and model size
- The ensemble of many trees reduces variance and improves generalisation, while a single tree is simpler, faster, and more interpretable but slightly less accurate in this experiment.
- Choose a decision tree when computation or interpretability is more important; choose Random Forest when predictive performance is the priority and resources are sufficient.

In practice, the "better" model depends on constraints: in offline batch diagnosis, the extra accuracy and robustness of the Random Forest may justify the higher cost, while on resource-limited or real-time systems a well-tuned Decision Tree or a smaller ensemble can be a more practical choice.

7. Learning objectives:

- Understand the key differences between Decision Trees and Random Forests in terms of structure, bias–variance, and interpretability.
- Measure and compare accuracy, training time, prediction time, and model size for both models on a real medical dataset.
- Explain how to choose between Decision Tree and Random Forest depending on accuracy needs and computational constraints

8. Reference:

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